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0178/03

October/November 2016

2 hours

Candidates answer on the Question Paper.

Additional Materials: Electronic Calculator
Geometrical Instruments
Tracing Paper (optional!)

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer all questions.

If working is needed for any question it must be shown below that question.

Electronic calculators should be used.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π use either your calculator value or 3.142.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 100.

This document consists of **14** printed pages and **2** blank pages.



- 1 The patterns show grey and white tiles.



pattern
number
1



pattern
number
2



pattern
number
3

pattern
number
4

- (a) Draw the 4th pattern in the space above.

[1]

- (b) State the name of the shape of the tiles used in the pattern.

Answer (b) [1]

- (c) Complete the table.

Pattern	1	2	3	4	5
Number of grey tiles	2	3	4		
Number of white tiles	2	4	6		

[2]

- (d) Find the total number of tiles in the 20th pattern.

Answer (d) [2]

- (e) Write an expression, in terms of n , for the number of grey tiles in the n th pattern.

Answer (e) [1]

- 2 (a) Simplify.

$$3(x + 2) - (2x + 7)$$

Answer (a) [2]

- (b) Make y the subject of the formula.

$$x = cy - 5$$

Answer (b) $y =$ [2]

- (c) Factorise completely.

$$ac - ad + bc - bd$$

Answer (c) [2]

- (d) Solve the following equations.

(i) $3x - 3 = 5x - 7$

Answer (d)(i) $x =$ [2]

(ii) $x^2 - 3x - 4 = 0$

Answer (d)(ii) $x =$ or $x =$ [3]



- 3 (a) Mpho and Lineo share M320 in the ratio 5 : 3.

How much more does Mpho receive than Lineo?

Answer (a) M [3]

- (b) Mpho spends $\frac{3}{5}$ of the money she receives.

- (i) Calculate how much she spends.

Answer (b)(i) M [2]

- (ii) Chocolates cost M11 each.

Work out the largest number of chocolates she can buy.

Answer (b)(ii) [2]

- (c) Ramasimong invests M600 at 6% per year compound interest.

Calculate the total amount he has after 3 years.

Answer (c) M [3]

- 4 (a) The basic working week for a chauffeur is 40 hours at a basic rate of M42.50 per hour. If he works overtime he is paid M15.50 per hour more than the basic rate.

Calculate

- (i) his wage for a basic week,

Answer (a)(i) M [1]

- (ii) his total wage for a week in which he worked for 45 hours,

Answer (a)(ii) M [2]

- (iii) the number of hours he worked in one week if he was paid a total wage of M2 338.

Answer (a)(iii) [3]

- (b) An insect covers 12 mm per step.

- (i) Calculate the number of steps needed to cover 2.4 m.

Answer (b)(i) [2]

- (ii) If it takes 5×10^{-2} seconds for each step, how long does it take to travel 3 m?

Answer (b)(ii) s [2]



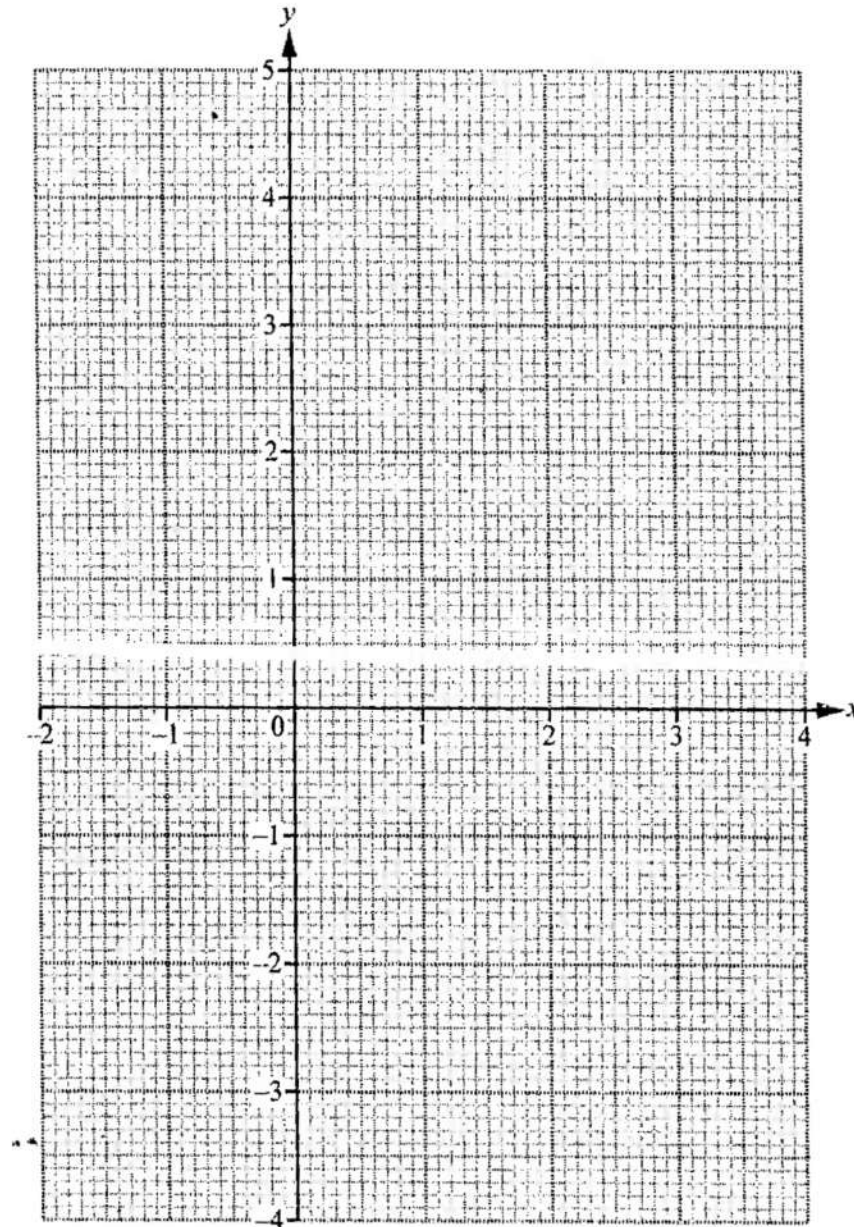
- 5 (a) Complete the table of the values for $y = x^2 - 2x - 3$.

x	-2	-1	0	1	2	3	4
y			-3	-4			5

[2]

- (b) On the grid, draw the smooth curve of $y = x^2 - 2x - 3$, for $-2 \leq x \leq 4$.

[4]



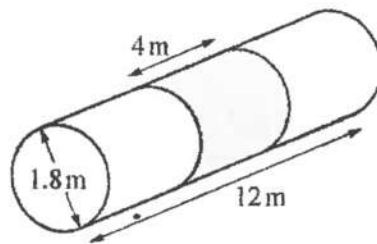
- (c) (i) On the same grid, draw the line $y = 2$.

[1]

- (ii) Write the x -coordinates of the points where the two graphs intersect.

Answer (c)(ii) $x = \dots\dots\dots$, $x = \dots\dots\dots$ [2]

- 6 The diagram shows a closed cylindrical petrol tank with the painted area represented by the shaded part. The tank has a diameter of 1.8 m and a length of 12 m. The length of the shaded part is 4 m.



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- (a) Sketch a net of the tank.

[2]

- (b) Calculate the area of the circular end of the tank.

Answer (b) m^2 [2]

- (c) Calculate the volume of the tank.

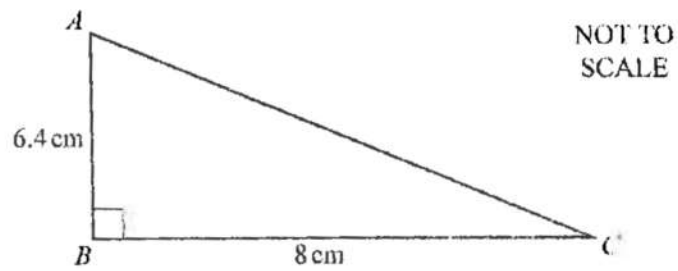
Answer (c) m^3 [2]

- (d) Calculate the area of the curved **unpainted** part of the tank.

Answer (d) m^2 [3]



- 7 (a) The diagram shows triangle ABC .
Angle $ABC = 90^\circ$, $AB = 6.4$ cm and $BC = 8$ cm.



Calculate

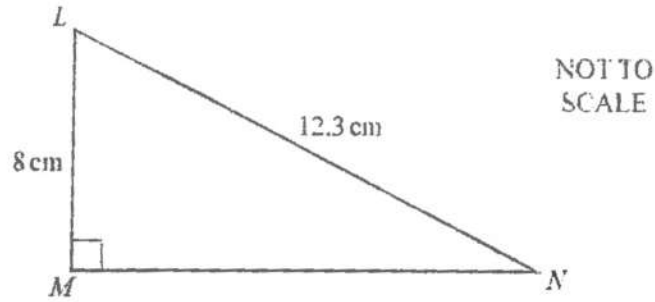
- (i) AC ,

Answer (a)(i) cm [2]

- (ii) angle CAB ,

Answer (a)(ii) [2]

- (b) In triangle LMN , angle $LMN = 90^\circ$, $LM = 8\text{ cm}$ and $LN = 12.3\text{ cm}$.



- (i) Calculate angle MLN .

Answer (b)(i) [2]

- (ii) Given that L is due north of M , calculate the bearing of L from N .

Answer (b)(ii) [3]

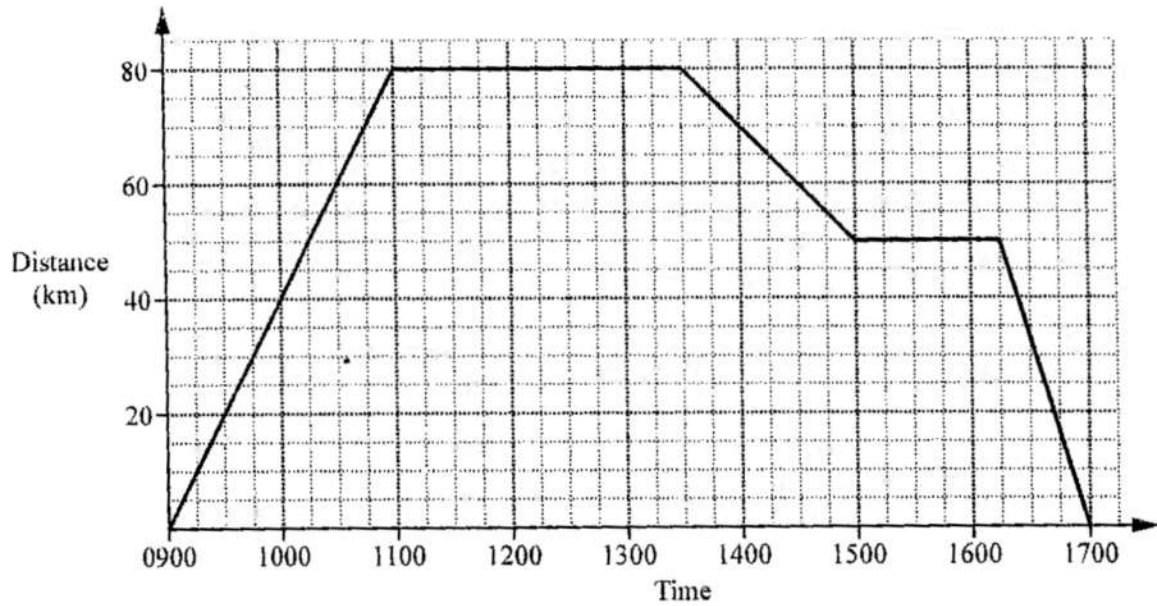
- (iii) MN is extended and the point P lies on this extended line so that $LP = 17.6\text{ cm}$.

Find NP .

Answer (b)(iii) cm [4]



- 8 The graph shows a bus travelling from Maseru to Leribe and back to Maseru.



- (a) Find the distance from Maseru to Leribe.

Answer (a) km [1]

- (b) How long does the bus stay at Leribe?
Give your answer in hours and minutes.

Answer (b) h min [2]

- (c) What happened at 1500?

Answer (c) [1]

- (d) (i) Between which times is the bus travelling fastest?

Answer (d)(i) and [1]

- (ii) At what speed is the bus travelling on this part of the journey?

Answer (d)(ii) km/h [2]

- (e) Calculate the average speed for the journey from Leribe to Maseru.

Answer (e) km/h [2]



- 9 (a) The table shows the temperatures at Litau in winter for one week.

Day	Mon	Tue	Wed	Thur	Fri	Sat	Sun
Temperature ($^{\circ}\text{C}$)	3	1	-2	-9	-1	4	7

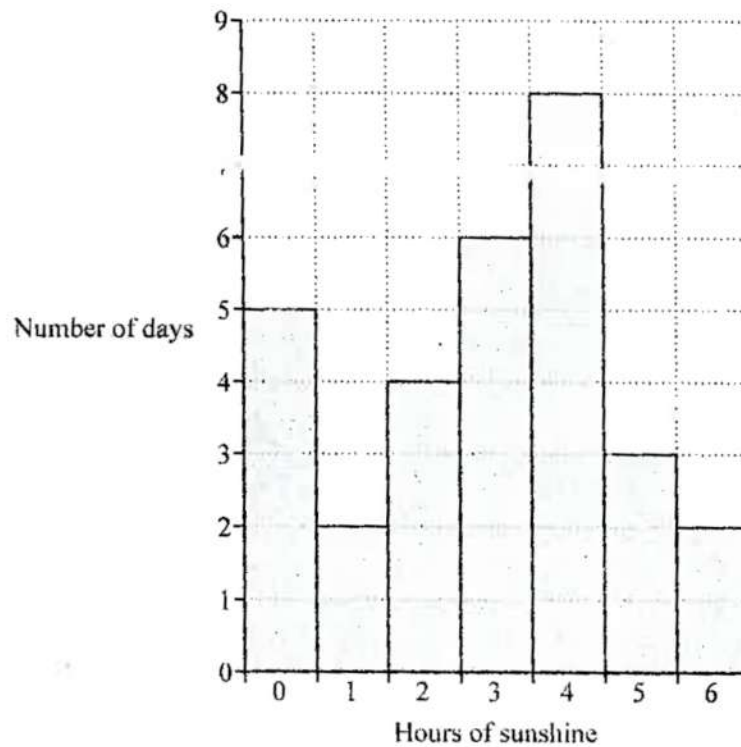
- (i) Write down the coldest day.

Answer (a)(i) [1]

- (ii) Calculate the mean temperature for the week.

Answer (a)(ii)..... $^{\circ}\text{C}$ [2]

- (b) The number of hours of sunshine at Litau was also recorded each day during one month. The results are displayed on the bar chart below.



- (i) Find the mode.

Answer (b)(i) h [1]

- (ii) How many days are there in this month?

Answer (b)(ii) [2]

- (iii) Complete the frequency table from the bar chart.

Number of hours of sunshine	
Number of days	

[2]

- (iv) Find the median.

Answer (b)(iv) h [1]

- (v) If the information on the bar chart is to be represented on a pie chart, find the angle of the sector representing 3 hours of sunshine.

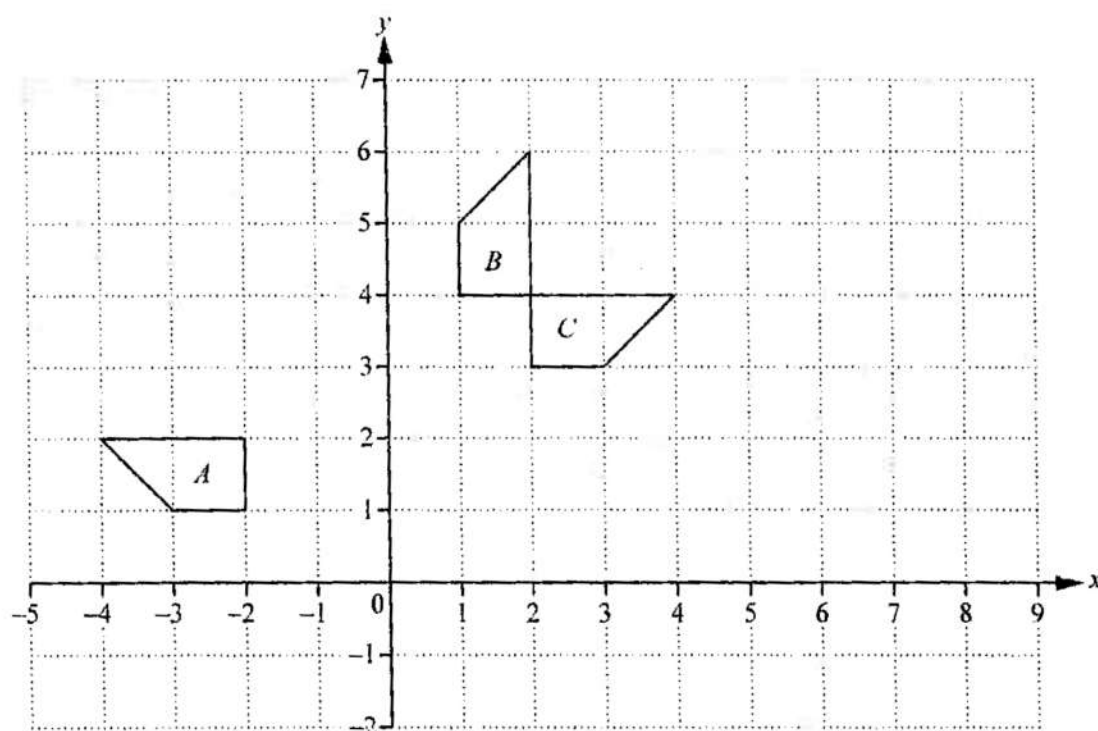
Answer (b)(v) [2]

- (vi) What is the probability that a day, chosen at random, has more than 4 hours of sunshine?
Give your answer as a fraction in its simplest form.

Answer (b)(vi) [2]



- 10 The diagram shows three trapeziums A , B and C .



- (a) Trapezium A is mapped onto trapezium D by translation $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$.

Draw and label trapezium D .

[2]

- (b) Describe fully the **single** transformation that maps

- (i) B onto C ,

Answer (b)(i)

[2]

- (ii) A onto B .

Answer (b)(ii)

[3]

- (c) Trapezium C is mapped onto trapezium E by an enlargement, scale factor 2, centre $(0, 2)$.

Draw and label trapezium E .

[2]